Experiment No.: 4

Date of Performance :
Date of Submission :

Aim: Wi-Fi network scanning using mobile apps (SSID, RSSI, channel mapping).

Required Equipment / Software :

- Android smartphone with Wi-Fi capability.
- Wi-Fi router/access point or access to nearby Wi-Fi networks.
- Wi-Fi analyzer/WiFi man scanning application such as WiFi Analyzer or NetSpot WiFi Analyzer (Android), as permitted by the laboratory.

Theory :

Wi-Fi is a wireless local area networking technology based on the IEEE 802.11 family of standards. A Wi-Fi scanner detects wireless access points within the range of the mobile device and displays parameters such as the Service Set Identifier (SSID), Received Signal Strength Indicator (RSSI), channel number, frequency band and sometimes security information.

SSID (Service Set Identifier): The SSID is the name used to identify a Wi-Fi network. It is normally displayed when a user searches for available wireless networks.

RSSI (Received Signal Strength Indicator): RSSI represents the received signal level at the mobile device. It is commonly shown in dBm. Values closer to 0 dBm indicate a stronger received signal; for example, −40 dBm is stronger than −75 dBm. In practical Wi-Fi measurements, approximately −30 to −50 dBm is very strong, −50 to −67 dBm is generally good, −67 to −75 dBm is moderate, and values below about −75 dBm are relatively weak. These ranges are approximate and can vary with device and application.

Wi-Fi Channel: Wireless networks operate on specific radio-frequency channels. In the 2.4 GHz band, channels are closely spaced and neighboring-channel overlap can cause interference. For typical 20 MHz Wi-Fi operation, channels 1, 6 and 11 are commonly selected because they are non-overlapping in many regulatory environments. The 5 GHz and 6 GHz bands provide additional channels, subject to device and regional regulatory support.

Channel Mapping: Channel mapping means listing or visually representing nearby networks according to their operating channels. It helps identify crowded channels and can assist in selecting a less congested channel for a Wi-Fi access point.

Procedure :

1. Switch ON Wi-Fi on the smartphone. Enable location permission if the mobile operating system requires it for Wi-Fi scanning.

2. Install and open a Wi-Fi analyzer/scanner application.

3. Allow the required permissions requested by the application.
4. Scan the surrounding area for available Wi-Fi networks.

5. Observe the SSID, RSSI, channel number and frequency band of each detected network.

6. Select at least 5 nearby networks, if available, and record their parameters in the observation table.

7. Repeat the scan at another location or at a different distance from the selected access point, if required by the instructor.

8. Group the detected networks according to their Wi-Fi channel and identify channels with high and low network occupancy.

9. Compare RSSI values. Remember that a less-negative RSSI value represents a stronger signal.

10. Prepare a simple channel map showing SSIDs against their respective channels and note the strongest/weakest signals.

11. Close the application after completing the measurements.

Observation Table :

Sr. No.	SSID	Band	Channel No.	Channel Width (MHz)	RSSI (dBm)	Signal Quality
1	Airtel_22	2.4	6	20	-73	Moderate
2	Airtel_Shardul07	2.4	1	20	-82	Weak
3	Ruhaani_4G	2.4	6	20	-86	Weak
4	TP-Link_9A22	2.4	11	40	-70	Moderate
5	Atharvaj_5G	5	157	80	-64	Good

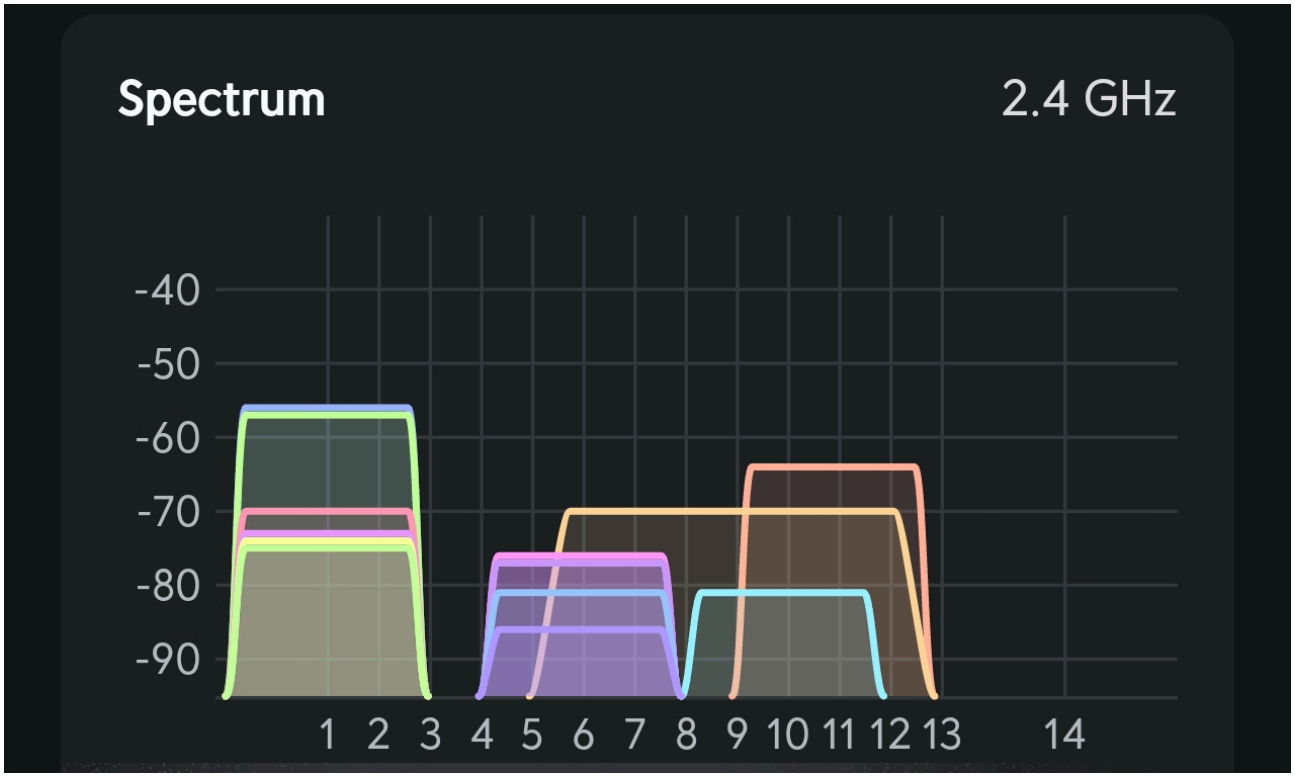
Suggested signal-quality classification: Very Strong (about −30 to −50 dBm), Good (about −50 to −67 dBm), Moderate (about −67 to −75 dBm), Weak (below about −75 dBm).

Channel Mapping Table :

Channel No.	Number of Networks Detected	SSID(s)	Highest RSSI (dBm)
157	**2**	**Atharvaj_4G, Atharvaj_5G**	**-61**
11	**2**	**TP-Link_9A22, Airtel_R k bharate**	**-65**
6	**4**	**Sakhadeos, Airtel_22, Ruhaani_4G, Airtel_Swam_2337**	**-77**

Graph :

Attach graph of spectrum of WiFi networks scanned using a mobile WiFi analyzer application.



Result :

Nearby Wi-Fi networks were successfully scanned using a mobile Wi-Fi analyzer application. The SSID, RSSI, operating channel and frequency band of the detected networks were recorded, and a channel map was prepared to identify channel occupancy and comparatively stronger or weaker signals.

Conclusion:

The experiment demonstrates how a smartphone can be used as a basic Wi-Fi network analysis tool. By studying SSID, RSSI and channel information, students can understand wireless network visibility, signal-strength variation and channel congestion. The observations also show the importance of selecting appropriate Wi-Fi channels to reduce interference and improve network performance.

Viva Voce Questions :

1. What is Wi-Fi?
2. What does SSID stand for?
3. What is RSSI?
4. Which is stronger: −45 dBm or −75 dBm?
5. Does high RSSI always guarantee high Internet speed ?

MARKS & SIGNATURE :

R1 (3 Marks)	R2 (5 Marks)	R3 (4 Marks)	R4 (3 Marks)	Total (15 Marks)	Signature